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Clutter Filtering for Diagnostic Ultrasound Color Flow Imaging

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Clutter filtering plays an important role in constructing a quality color flow map in ultrasound Doppler imaging. Signals from slow-moving tissues and vessel walls are clutter as they often mix with reflections from blood and should be suppressed for the further correct estimation of flow parameters. Their complete suppression in color flow imaging is difficult, because these signals on average are 40-60 dB more powerful than the signals from blood, the length of the Doppler sequence is very short, and there is always a demand for a real-time operation. This article provides a general model of the Doppler signal and discusses filters based on polynomial and adaptive regression, empirical mode decomposition, and prospective combined approaches to blood flow filtering.

Introduction

Color flow imaging (CFI) is a Doppler mode for ultrasound medical diagnostic devices allowing the dynamics and spatial distribution of the blood flow to be observed. It provides opportunity to follow the development of stenosis, tumors, and aneurysms, to assess the effectiveness of bypass, and to determine the quality of heart valve operations. The CFI mode is often used for study of blood flow that has a low velocity relative to the surrounding tissues [1, 2]. An example of such use is the diagnosis of the microcirculation in the eyeball, studies of the development of microvessels supplying tumors, in chemotherapy, analysis of heart muscle functioning, etc. An advantage of the CFI mode over other noninvasive diagnostic methods such as computed and magnetic resonance angiography is the ability to carry out real-time investigations. In addition, ultrasound machines are very accessible, universal, and available in many therapeutic institutions.

Given the widespread use of ultrasound medical devices, it is very important that the CFI mode gives high-quality diagnostic information. Errors in blood flow visualization can lead to incorrect diagnoses and complications during the long-term follow-up of patients [3]. From the signal processing point of view, there are two main features of the data affecting the quality of CFI operation [4]. First, the signal contains not only blood flow components, but also clutter, which is a reflection from slow-moving tissues mixed with flow signals. Clutter can be generated by the pulsation of artery walls, the heart, shaking of the doctor's hands, intestinal peristalsis, etc. Its power is generally 40-60 dB greater than that of blood flow. Secondly, the length of the CFI sequence should always be relatively small (usually no more than 5-17 samples) to allow real-time operation.

Processing of CFI data involves two main stages: filtering and velocity estimation in each element of resolution. The velocity estimation is generally performed using an autocorrelator to calculate the mean phase shift of the Doppler sequence [5]. It is assumed that only white noise and the blood flow signals arrive at the input of the autocorrelator, the tissue clutter being completely suppressed at the filtering stage. Therefore, incorrect filtering can lead to burst artifacts or incorrect display of blood flow on the screen.

In recent years, there has been a multitude of studies on the clutter filtering in CFI [6-18]. Classical infinite impulse response (IIR) and finite impulse response (FIR) filters are very suitable for analysis of fast blood flow [8]. However, in conditions in which blood flow velocity is

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low relative to tissue movements or when their spectra overlap, such filters are unable to provide complete suppression of tissue signals [18]; adaptive and regressive algorithms may be more effective in these situations.

Most of the well-known adaptive filters are designed on the basis of methods for tissue motion compensation and eigenvector decomposition. It is interesting to compare these with the algorithm for decomposition into empirical modes and a polynomial regression filter approximating the tissue signal with a polynomial curve, minimizing errors by using the least squares method.

The decomposition into empirical modes is the first step in the Gilbert–Huang transform [19-25]. It needs no *a priori* basis and is carried out adaptively obtaining the empirical mode functions directly from the input data.

In the filter based on the eigenvector decomposition, the orthogonal base is computed using a covariation matrix by Karhunen–Loève decomposition [6-14]. The component with the greatest amplitude corresponds to the signal from the tissues. Approximation is carried out by the least squares method, which gives the greatest level of tissue signal suppression for a filter of a given order. As in the method based on the empirical mode decomposition, the computation complexity is again too high, which hinders the introduction of these algorithms into devices operating in real time [8].

Doppler Signal Model

Processed CFI channel data can be presented as a four-dimensional matrix; one dimension corresponds to the frame No., another to the beam No., the third to the fast-time, i.e., depth, and the fourth to the slow-time, i.e., the number of pulse in the ensemble. Processing in the CFI mode addresses ensembles of slow-time; they can be described as a vector of K complex samples $x = [x_0, x_1, ..., x_{K-1}]^T$ taken from one point in space with a pulse repetition frequency. The ensemble of slow-time in the blood flow region usually consists of three components: 1) reflections from erythrocytes; 2) reflections from tissues; 3) thermal noise from the receiver device. If there is no blood flow at the point of interest, the ensemble includes only thermal noise and the tissue signal. The ensemble of slow-time can then be presented in two forms depending on the presence of blood flow:

$$x = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{K-1} \end{bmatrix} = \begin{cases} b + c + n & \text{(blood flow);} \\ c + n & \text{(tissue).} \end{cases} \quad (1)$$

In Eq. (1), x_k corresponds to the kth element of the ensemble, while a, b, c, and n are the vectors of length K corresponding to the reflections from erythrocytes, reflections from tissues, and the noise component [6, 10].

Components b and c come from different sources located in different points in space and are therefore statistically independent. The thermal noise is modeled as white noise due to its large bandwidth at the input of the Doppler processing channel [14]. The probability density $p_x(x)$ of slow-time ensemble is described as the centered complex Gaussian process:

$$p_x(x) = \frac{1}{\pi^K |R_x|} e^{-x^{*T} R_x^{-1} x}. \quad (2)$$

Being Gaussian, the ensemble is fully characterized statistically by a second-order moment, information on which is present in the correlation matrix:

$$R_x = E\{xx^{*T}\}, \quad (3)$$

where E is the mathematical expectation operator of dimensionality $K \times K$.

Given the statistical independence, this can be written as:

$$R_x = \begin{cases} R_c + R_b + R_n \\ R_c + R_n \end{cases} = \begin{cases} R_c + R_b + \sigma_n^2 I; \\ R_c + \sigma_n^2 I. \end{cases} \quad (4)$$

In Eq. (4), R_c is the clutter correlation matrix, R_b is the erythrocyte signal correlation matrix, σ_n^2 is the dispersion of thermal noise, and I is the unitary matrix of dimensionality $K \times K$.

Determination of the presence of blood flow components in the ensemble being analyzed is very important for the map formation. This requires suppression of the tissue reflection component whose power is usually more than 40 dB greater than that of the other components. This is the main task of the filtering stage, which in case of relatively small blood flow velocities is more complex due to the superposition of blood spectra with the clutter.

Filtering Algorithms

Filtering based on polynomial regression. In contrast to the IIR and FIR algorithms, which carry out filtering as a result of linear time-independent superposition of samples, the regression filters approximate the input signal with a polynomial in the time domain, and the classical analysis of the pulse and frequency characteristics is not applicable to the polynomial regression filters. The

clutter signals are generally slow and can be described by a first- or second-order Legendre polynomial [14, 26, 27]. With the least squares fitting method the Doppler sequence is projected onto a subspace W, which contains a set of polynomials $B = \{b_0(t), b_1(t), ..., b_i(t)\}$ such that $b_i(t) = c_{0i}, c_{1i}t, ..., c_{ii}t^i$. This projection of ensemble x_k is present in W and is described as

$$\hat{x}_k = \sum_{i=0}^{p} \alpha_i b_i(t_k), \tag{5}$$

where $t_k = kT_{PRF}$ is the time interval between the null and kth impulses in the packet; $k = 0, 1, ... K - 1$.

The residue $y = x - \hat{x}$ corresponds to the blood flow component not present in in the subspace W because of orthogonality to $\hat{x}$. The set of coefficients α is computed as

$$\alpha_i = \frac{(x, b_i)}{\|b_i\|^2} = \frac{\sum_{k=0}^{K-1} x_k b_i(t_k)}{\sum_{k=0}^{K-1} b_i^2(t_k)}, \; i = 0, 1, \ldots, p. \tag{6}$$

For the normalized basis, we can write that $\hat{x} = B^T A$ and $A = Bx$, while $\hat{x} = B^T Bx$, and the filtered signal can be expressed in the following form:

$$y = x - \hat{x} = (I - B^T B)x = Fx, \tag{7}$$

where F is the matrix of the polynomial filter. Thus, the difference equation is reminiscent of a FIR filter but is time-dependent:

$$y(l) = \sum_{n=0}^{N_D - 1} f(l, n) x(n), \; l = 0, 1, \ldots, N_D - 1. \tag{8}$$

Regression filters are adaptive in the sense that the coefficients of the polynomial are inserted beneath the ensemble [see Eq. (6)]. These filters do not require samples after filtering to be excluded and are also characterized by high attenuation in the suppression band but have an extended transition band which can lead to suppression of the blood flow components and an increase in the correlatedness of noise, expressed as the appearance of a burst artifact [11, 14].

Filtering based on adaptive regression. This approach consists of seeking unique basic functions which are eigenvectors of the correlation matrix of the initial data and are therefore adapted to the statistics of the clutter and blood signals. This is a completely adaptive approach, as not only the coefficients, but also the weights of the basis, are determined by the input data. This approach has different names: the Hotelling transform, the principal components method, and the discrete Karhunen–Loève transformation [14-17, 28-30]. In this method, the ensemble of slow-time is decomposed as follows:

$$x = \sum_{k=0}^{K-1} \gamma_k e_k; \; E\{\gamma_l \gamma_l\} = \begin{cases} \lambda_k, & k = l; \\ 0, & k \neq l, \end{cases} \tag{9}$$

where γ_k is the kth decomposition coefficient and e_k and λ_k are, respectively, the eigenvector and eigenvalue corresponding to the kth orthogonal basic function. The eigenvectors and eigenvalues are linked with the ensemble by

$$\sum_{k=0}^{K-1} \lambda_k e_k e_k^H = E\{xx^H\} = R_x. \tag{10}$$

The correlation matrix is found by grouping ensembles [6]. Matrix A of size $K \times M$ is formed from M ensembles and is used to compute the correlation matrix R_x of size $K \times K$:

$$A = \frac{1}{\sqrt{M}} \begin{bmatrix} | & | & & | \\ x_1 & x_2 & \cdots & x_M \\ | & | & & | \end{bmatrix}; \tag{11}$$

$$R_x = AA^H = \frac{1}{M} \sum_{m=1}^{M} x_m x_m^H. \tag{12}$$

Decomposition of matrix R_x gives K eigenvectors and eigenvalues. As the eigenvalues are arranged in decreasing order, the Karhunen–Loève transformation is optimal from the energy concentration point of view [3], as no other transformation places so much energy in the first components. This filtering procedure consists of excluding from the signal the components with the highest energy. The order of the filter is determined by the number of excluded components. At a given filter order, as compared with other algorithms, the Karhunen–Loève algorithm provides the best suppression of the tissue reflection component and gives the best approximation to the ensemble using the least squares method [10].

However, as spatial averaging is performed on formation of the correlation matrix, ensembles must have identical clutter statistics; ensembles within one beam or a group of beams should be taken to ensure stationarity on the fast-time coordinate [28]. Along the slow-time coordinate, conversely, stationarity is not obligatory, so the Karhunen–Loève algorithm can adapt even when tissue movement accelerates. The tissue reflection component has high amplitude and a narrow spectrum, so it can to a significant extent be approximated by a single eigenvector; thermal noise is distributed through the whole subspace of eigenvectors [11].

Empirical mode decomposition. The method proposed by Huang and termed empirical mode decomposition [22-25] is an adaptive iterative computational procedure for decomposition of initial data into empirical modes. This approach has not yet received strict mathematical grounding but has already proved itself useful in solving many practical tasks.

The empirical mode is an oscillation in which instead of constant amplitude and frequency, as in a harmonic signal, the amplitude and frequency can change within the ensemble. The empirical mode has the following properties: a) the numbers of minima, maxima, and nulls must not differ by more than unity; b) at each point of the function, the mean value determined via the upper and lower envelopes must be zero.

The essence of the empirical mode decomposition method consists of sequential computation of the functions of empirical modes c_j and residues r_j:

$$x_k = \sum_{j=1}^{m} c_j(t_k) + r_m(t_k), \tag{13}$$

where m is the number of empirical modes.

The process of seeking empirical modes consists of the following steps [19, 20]; 1) let $u_1(t) = x_k$; 2) we seek the local extremes $u_1(t_k)$ (if just one maximum or minimum is found, then we exit the cycle and take $u_1(t_k)$ to be the single empirical mode); 3) interpolation of the extremes is used to compute the upper $h(t_k)$ and lower $l(t_k)$ envelopes; 4) the local mean is determined: $\mu_1(t_k) = [h(t_k) + l(t_k)]/2$; 5) the mean is subtracted: $u_1(t_k) = u_1(t_k) - \mu_1(t_k)$; 6) we obtain the mean square deviation (MSD) from $u_1(t_k)$; 7) if the MSD is below the threshold, then $u_1(t_k)$ is taken as the empirical mode: $c_1(t_k) = u_1(t_k)$, otherwise the algorithm goes back to step 2.

This approach is convenient, as the decomposition forms a set of functions ordered by frequency [21]. This can be restricted to the first mode, as it has the highest frequency: it corresponds to the blood flow component. The filtering procedure then consists of seeking the first mode, which is then taken as the filtered signal.

The ensembles are characterized by the tissue reflection component having high amplitude and the blood flow component being relatively slow, which not infrequently leads to exit from the cycle at algorithm step 2. A modification is therefore introduced to the first cycle, replacing exit from the cycle by including the first and last elements of the ensemble in the number of extremes.

A combined filtering scheme. This filtering scheme includes an algorithm to compensate for tissue movements, a polymodal regression filter, and recursive linear filtering used to capture data and suppress random components in the set of correlation coefficients used for computation of the blood flow velocity and correlation masks allowing burst artifacts to be decreased.

The tissue movement compensation algorithm operates with the phase of the ensemble [9]. It acts on the assumption that before filtering, the mean phase shift of the ensemble is determined by the speed of the tissue reflection component. If the phase of the ensemble is substituted by the mean phase shift, suppression of the tissue reflection components can use a standard high-frequency filter [7]. We used a polynomial regression algorithm, a special case of which is the filter subtracting the mean, equivalent to subtraction of its constant component from the ensemble:

$$y = (I - E\{Dx\})Dx = FDx; \tag{14}$$

$$D = \mathrm{diag}[e^{-j0}, e^{-j\varphi}, \ldots, e^{-j\varphi(K-1)}]; \tag{15}$$

$$\varphi = \arg \sum_{k=0}^{K-2} x_k x_{k+1}^{*}, \tag{16}$$

where F is the matrix of the mean subtraction filter and D is the operator for tissue movement compensation.

The mean subtraction filter was selected, as it is simple to use, does not lead to losses of samples, and is economic in terms of computational resources, as compared with the classical IIR and FIR algorithms.

Then, correlation coefficients are calculated and blurred by a biexponential filter, which helps reduce burst artifacts and provide a stable reflection of blood flow. The biexponential filter is a second-order filter equivalent to two first-order filters [14], when the value y_n is put into the second filter in place of x_n:

$$y_n = \alpha x_n + (1 - \alpha) y_{n-1}; \tag{17}$$

$$\alpha = \min\left[1, \frac{|x_n - y_{n-1}|}{\Xi}\right], \tag{18}$$

where α is the coefficient of filtering and Ξ is the filter tuning parameter characterizing the smoothing value.

To adapt the algorithm to tissue movement, phase φ is averaged for the group of beams with identical statistical characteristics [14].

As none of the steps is expensive in terms of resources, the filtering scheme proposed here not only allows computational economy as compared with the adaptive and classical algorithms discussed here, but also wins in terms of quality as assessed in terms of the absence of distortion to the blood flow profile and the ROC-analysis.

Conclusions

This article provides a review of filtering methods for the Doppler ultrasound color flow imaging. The algorithms discussed here can be used in both 2D and 3D/4D US systems. Development of an optimum algorithm for blood flow filtering is a laborious task, solution of which requires finding a compromise between high frame rates and imaging quality, which is especially important in real-time volume visualization [31].

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